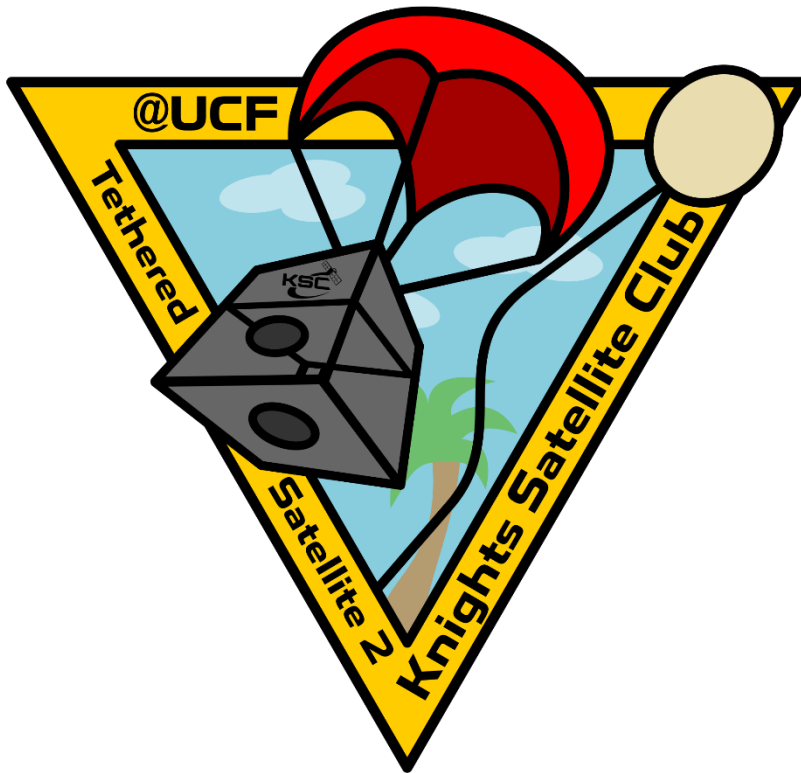




Preliminary Design Review (PDR)

Tethered Satellite 2 Mission A

Knight's Satellite Club

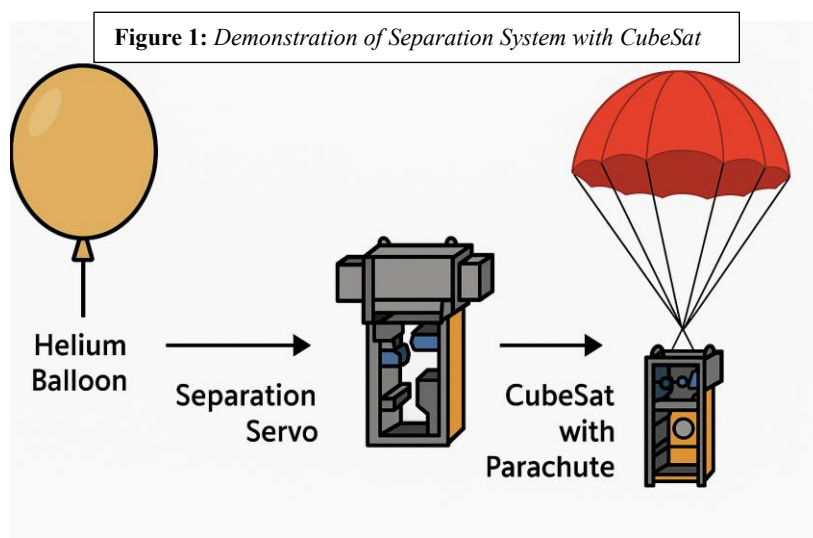


Introduction

Through previous missions of the tethered satellite(T-Sat) projects including T-Sat-0, T-Sat-1 and now T-Sat-2 the focus has and will remain to create a beginning friendly platform that can progress into a way for concurrent members or new members to get hands-on-experience ranging from the mechanical, software, electrical and everything in-between while gaining deeper knowledge involving CubeSats. Specifically in T-Sat-0, with it being the first iteration, there were more experienced members that worked on creating the groundwork, which involved many trials and errors. Which would later allow for a smooth process for future missions. The scope of the mission involving T-Sat-0 began with creating a 1U CubeSat that would be able to be deployed through a parachute through a spring-loaded system while taking pictures of the flight around every ten seconds. Which would separate from the helium balloon through a controlled burn-wire system. Where the helium balloon is what allows the Knight's Satellite Club to have high-altitude flight still in University of Central Florida (UCF) campus. As such, an incorporation to multi-core software had to be created. Allowing for live telemetry to the ground station team at the landing sight. With T-Sat-0 being the first iteration there are bound to be some errors to show up during and after the launch which was shown through the faulty accelerometer measurements, caused by the difficulty winding weather conditions of the launch date.

As such T-Sat-1 became to flourish. The goal of this mission was not to introduce a new experiment but to refine and prove the operations of T-Sat-0 are replicable in a shorter timeframe. Since T-Sat-0 was a proof of concept, T-Sat-1 became a demonstration of maturity. The most crucial point being the removal of the accelerometer used for freefall detection. Since past testing it would cause premature parachute deployments. With a focus that appeared through the testing of T-Sat-1 being how random the burn wire would or would not work when operated to do so. Which was fixed later on in time for the launch date. Proven by the data collected the day of the launch, allowing the T-Sat-1 team to have a satisfying conclusion to their mission.

After leaving such a legacy of previous T-Sat missions, T-Sat-2 has some big shoes to fill on both missions A and B, with mission A having no intentions of disappointing. Since mission A tethered satellite project involves staying near our roots and continuing using a deployment system except with a catch. With our focus being on a separation system between the top panel of the CubeSat and the rest of the housing with safe landing through a parachute.



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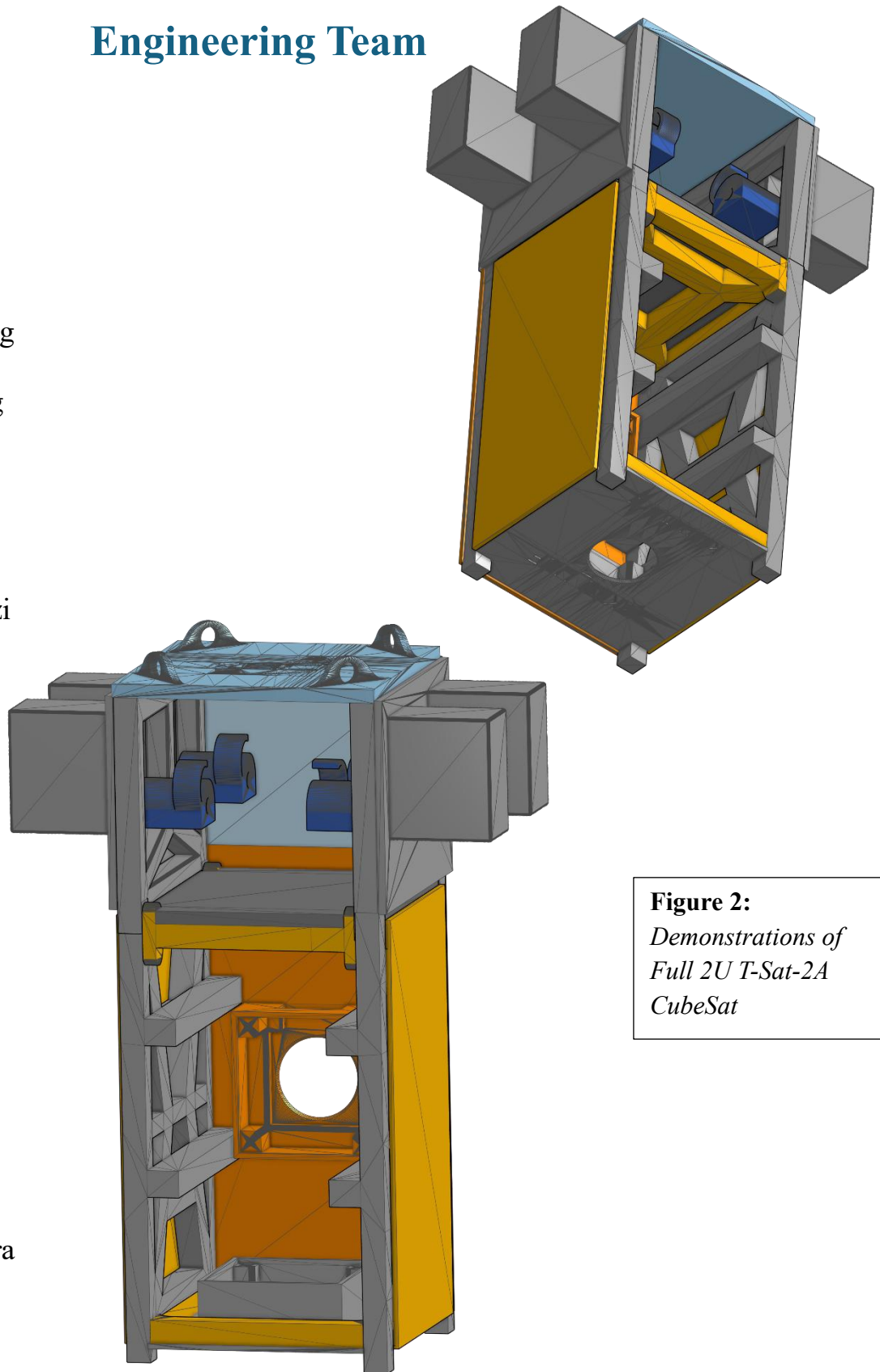


Figure 2:
*Demonstrations of
Full 2U T-Sat-2A
CubeSat*

Mission Evolution

Though T-Sat being a focus on learning about the vast world of CubeSats. In where many space graded CubeSat involve separation systems, it was a no-brainer to involve one for this mission. With the creation of a separation system what once was a single system, turns into a two-layer system with specifically having a size of a 2U CubeSat, compared to the previous missions only being 1U. As stated, the two-layer system involved would be separated at the top of the 2U CubeSat, since most of the CubeSat will stay intact expect the top removal panel. In doing so, the top panel becomes system two. With only part of system two that will physical contact with the CubeSat being the hooks and top panel. The hooks being used to have a linkage to allow a closed CubeSat until commanded to separate. While the rest of system two would be used to intertwine the rope to the helium balloon that will allow for elevation in T-Sat-2's launch. As shown in figure 3 and 4, the hardness points for the rope.



Figure 3: System 2 (Top Panel of T-Sat-2A)

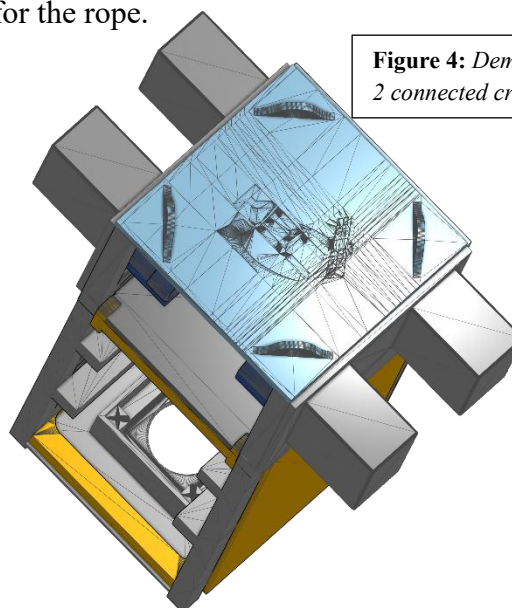


Figure 4: Demonstration of System 1 and 2 connected creating complete Housing.

With the point of intent of system two being used as the connecting point for the helium balloon and CubeSat. It leaves system one with all the strain involving the electrical, software, and hardware. With system one having a size of a 2U CubeSat, the decision involving how the space must be used has to have a bit of creativity. Starting with the first tier, it involves having a dual camera set-up. Where one of the cameras will be facing directly forward by the bottom panel, with the second camera being mounted on the side panel. As was done before in T-Sat-0&1. Right in between both camera mounts, the indentation for the mounts that will be used for the PCB sled, that contains a triple layered PCB stack. Once filled a top panel will need to mount through threaded inserts and bolts. To allow a flat surface the parachute can be deployed from. With the side panels on the top of the 2U Cub). With containing indents that are going to be shown from the outside of the CubeSat so there can be a fitting for the servos that will oversee creating a spin to unlock from the top panel (Figure 4). With each perspective wall containing a

form of trusses to allow for swaying of the weight attached of the panels not create any strain or bend.

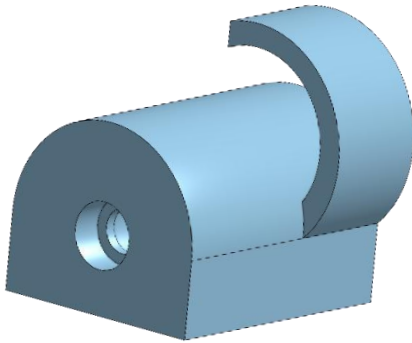


Figure 5: *Servomotor Attachment*

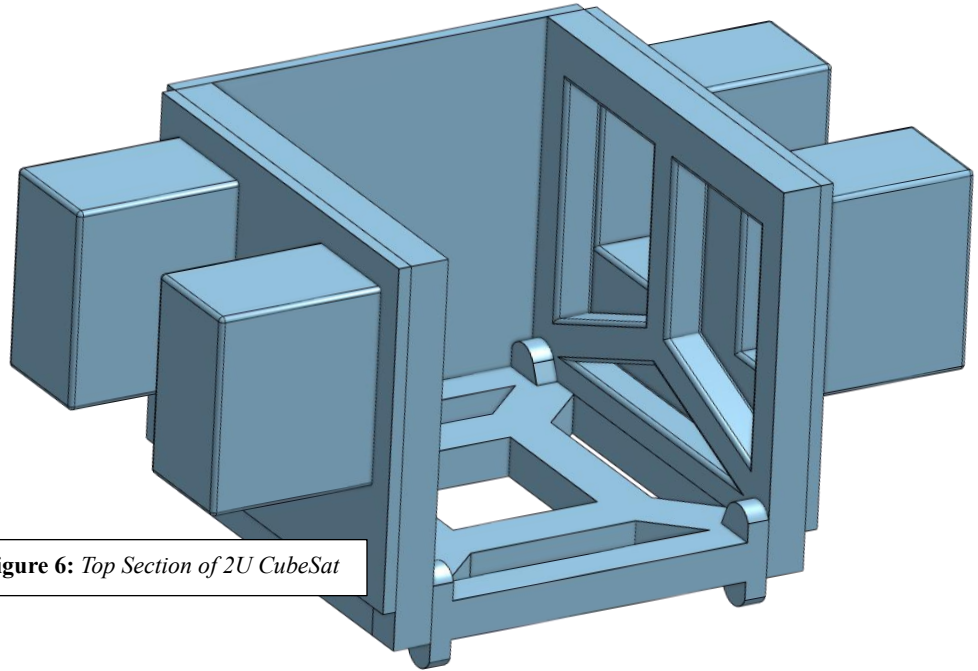


Figure 6: *Top Section of 2U CubeSat*

The creation of the housing and separation system for T-Sat-2A would not be able to operate without either the electronics or software. In doing so, this project will use an ESP32 as the operating microcontroller. Where the software team as viewed the past T-Sat missions C+ code for inspiration when creating the test code for when doing test on either the sensors or camera. While integrating the code that seems to have been debugged into the final version for launch. In which will contain five stages in the Entry, Descent, and Landing (EDL) sequence.

The first stage, **Calibration**, begins once the ready-to-launch pin is removed, enabling power distribution and command flow through all onboard modules. This step serves as the initial system activation; if the pin remains in place, no electrical or mechanical actions occur within the CubeSat. Following calibration, the **Preflight** phase serves as a waiting condition in which the system continuously monitors altitude readings. Once again in altitude is detected by the onboard sensors, the CubeSat automatically transitions to the next operational state. During the **Ascend** phase, the altimeter becomes active, providing real-time altitude data to the ground team at the launch site. Throughout this ascent, both cameras capture images, and all other onboard systems remain powered and on standby, awaiting the next sequence trigger. Upon reaching the predetermined altitude, the **Freefall** phase is initiated. The servomotors receive a command signal to actuate the separation mechanism, releasing the lower 2U CubeSat from the upper panel assembly. Once detached, the CubeSat begins its controlled freefall, during which the parachute deploys through aerodynamic drag. Finally, during the **Landing** phase, the parachute fully inflates, rapidly slowing the CubeSat's descent. This stage occurs almost simultaneously with the start of freefall. As the system descends safely to the ground. All data

including altitude readings and captured images, continuing to be logged to the onboard SD card for post-flight analysis.

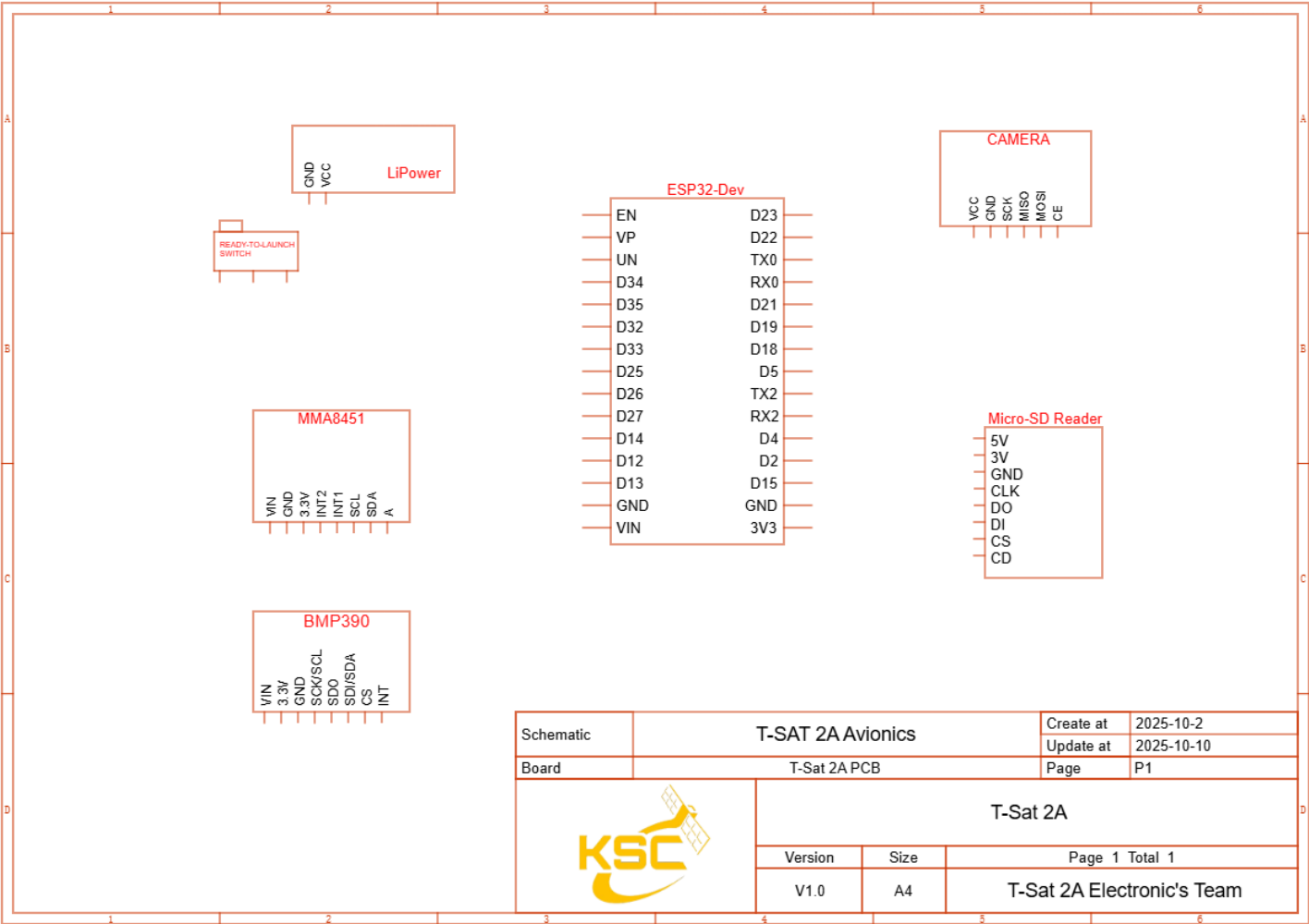


Figure 7: 2D Schematic of the Electronic Components for T-Sat-2A

Through the electrical works needed for this T-Sat-2 mission A, will involve reused past ideas from T-Sat-0. Through the previous finds the involved components range from an ESP32, altitude sensor (BMP390), accelerator (MMA8451), micro-SD reader, camera module inserts, and power converter connected to a switch. With the ESP32 being used as the communication device to rest of modules. With the accelerator module though being featured in figure 7 will not be taking place in the finial form of the PCB to not increase complexity of code, since through past showing of T-Sat missions has shown faultiness.

With all the modules being placed into one single PCB. It was found that multiple connections needed to operate both cameras and servos, would still need communication from the ESP32. As such there will be a second PCB that will be used as a wiring board. In doing so

there is more area for connecting inserts that the camera and the servomotors will still be communicating with the ESP32. With the final layer of the PCB stack being a 3-D printed panel where the battery can stay mounted through the PCB sled. The PCB sled being a 3-D printed housing that will stay supported through the railing created by the sidewall. Allowing a cube format where all the PCBs can mount vertically, specifically for space savings inside the 2.5U CubeSat.

Bill of Materials

Part Name/Item Name	Date Issued	Part Number	Description	Quantity Requested	Unit Price (USD)	Subtotal (USD)
36" Nylon Parachute	10/1/2025	#29095	Apogee Nylon Parachute	1	\$18.69	\$27.78
Black PETG Filament	10/1/2025	ASIN: B0DS4CR7M1	DEEPLIFE Rapid PETG Filament	2	\$9.89	\$21.14
Ready-to-launch Switch	10/1/2025	#SW244-ND	SS-5GL2D Switch	2	\$2.04	\$9.36
Boost Converter	10/1/2025	PRT-10255	LiPower Boost Converter	1	\$17.50	\$26.68
4 Pack 25KG Digital RC Servos	10/1/2025	ASIN: B0D6RFFLPK	25KG Digital RC Servomotors	1	\$51.29	\$54.88
MicroSD Module	10/1/2025	Product ID: 254	Adafruit MicroSD Breakout Board	1	\$7.50	\$13.09
2 Pack MicroSD Cards	10/1/2025	ASIN: B07K81Z6DF	Micro Center 32GB Class 10 MicroSDHC	1	\$9.19	\$9.83
Adafruit BMP388 Altimeter Module	10/1/2025	Product ID: 3966	Adafruit BMP388 Altimeter - STEMMA QT	1	\$9.95	\$15.71
7.4V Battery Pack	10/1/2025	#09225	7.4v 260mAh 2-Cell LiPo Battery (1/pk)	1	\$11.94	\$20.56
JLCPCB Maker	10/1/2025	N/A	JLCPCB Website	N/A	~\$10	\$0.00
3 Pack ESP32 Dev Boards	10/1/2025	ASIN: B0D8T53CQ5	ELEGOO ESP-32 Dev Board USB-C 2.4GHz Dual Mode	1	\$16.99	\$18.18
Arducam Mega SPI Camera	10/1/2025	ASIN: B0BW4L21KS	Arducam Mega SPI Camera 3MP Fixed Focus	2	\$25.99	\$55.62
Step Down Buck Converter	10/1/2025	ASIN: B0819Y8PJZ	DC-DC 6V-40V to 1.2V-35V Step Down Converter	1	\$16.99	\$18.18
BMP390 Single Sensor	10/18/2025	828-BMP390TR-ND	BMP390 Altimeter Sensor Pin	1	\$3.15	\$8.36
					Total:	\$299.37

Figure 8: *Proposal of all Materials Needed. (Prices may vary)*

With budget in mind through this project the team has accounted for a bill of materials (Figure 8) need for this T-Sat mission to succeed. With most of the budget needed for modules that will be integrated into the PCBs. Such as the two cameras needed allocation for one-fifth of the budget. With another huge chunk of the budget allocating for the four servomotors adding another one-fifth of the budget. To create security of the budget that will be allocated for this T-Sat mission the team calculated each component with its own pricing including shipping and taxes, creating an inflated total price. With the most up to date insert to the budget being the BMP390 sensor that is just an altitude sensor unlike the past one that was accounted for that had the whole module. This new sensor would just be a single pin like sensor that would be soldered into the PCB itself with needs of adding copastors and resistors to create a working sensor. As such it has not been fully decided if the team will adjust into the new sensor.

References

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2. *NASA Procedural Requirements 7123.1C – Systems Engineering Processes and Requirements*. National Aeronautics and Space Administration, Office of the Chief Engineer, Washington, D.C. Accessed via NASA Directives and Regulations System (NODIS): https://nodis3.gsfc.nasa.gov/displayCA.cfm?Internal_ID=N_PR_7123_0001_&page_name=AppendixG
3. *Knight's Satellite Club – UCF Wiki*. University of Central Florida, Knight Space Club, Orlando, FL. Available at: <https://wiki.kscucf.org/>
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5. *SEDS Project 39-B Critical Design Review – Final Draft v2*. Students for the Exploration and Development of Space (SEDS), University of Central Florida, 2022. (Uploaded reference document used for subsystem documentation style and presentation layout.)
6. *T-Sat-2 Systems Engineering Handbook – Draft Template*. Knight Space Club, University of Central Florida, 2025. (Internal document prepared by A. Salazar; used for subsystem formatting and requirement structure.)
7. *T-Sat-2 Project Timeline*. Knight Space Club, University of Central Florida, 2025. (Internal scheduling reference for project phase mapping and milestone planning.)
8. *T-Sat-2A Mission Concept Review (MCR)*. Knight Space Club, University of Central Florida, 2025. (Primary source for mission objectives, system description, and EDL sequence definition.)